

HealthConnect Clinic Experience Lab

Week 7 Final Analytical Report & Validated Performance Metrics

Project Scope: Comprehensive validation of appointment metrics, operational distributions, and Streamlit dashboard production for clinic optimization.

1. Executive Summary

This report encapsulates the final validated findings from the HealthConnect operational dataset (N = 5,000). Through automated data hygiene protocols, Python-driven audit scripts, and interactive Streamlit deployment, all core hypotheses regarding patient attendance, no-show risks, and clinic resource utilization have been rigorously verified.

2. Validated Core Performance Metrics (KPIs)

Metric Category	Validated Benchmark	Operational Significance
Total Sample Volume	5,000 Appointments	Provides high statistical significance across clinical workflows.
Attendance Rate	46.28%	Indicates less than half of booked patients successfully complete visits.
No-Show Rate	48.46%	Primary bottleneck: Nearly half result in unnotified absences.
Cancellation Rate	5.26%	Minor administrative or patient-initiated schedule drops.
Mean Lead Time	29.64 Days	Long booking windows correlate with higher no-show probability.
Average Waiting Time	24.19 Minutes	Critical efficiency metric for on-site clinic capacity planning.

3. Data Quality & Audit Status

Validation Test	Target Field	Expected Result	Status
Primary Key Uniqueness	appointment_id	100% Unique	PASSED
Duplicate Record Scan	Entire DataFrame	0 Duplicate Rows	PASSED
Missing Value Isolation	Operational Fields	Strictly Dict-Aligned	PASSED

4. Strategic Recommendations

Lead Time Mitigation

Appointments booked greater than 14 days out show elevated no-show risk. Implement automated multi-channel SMS reminders at 7 days and 24 hours prior.

Dashboard Operationalization

Transitioned successfully to a dynamic Streamlit application (`app.py`), granting real-time filtering and visual clarity for operational stakeholders.